Career Launcher MBA

QA Algebra-06

Greatest Integer Function; Maxima and Minima

Question 1

If the symbol [x] denotes the largest integer less than or equal to x, then the value of

$$\left[\frac{1}{4}\right]+\left[\frac{1}{4}+\frac{1}{50}\right]+\left[\frac{1}{4}+\frac{2}{50}\right]+\ldots\left[\frac{1}{4}+\frac{40}{50}\right] \text{ is}$$

(1) 40

(2) 28

(3) 3

(4) 0

Question 2

The number of solutions of $[x] = x + 1$ is

(1) 0

(2) 1

(3) 2

(4) infinite

Question 3

Find how many positive real values of x satisfy the equation $2[x]^2 = 5x + 2$, where [x] denotes greatest integer less than or equal to x.

(1) 0

(2) 1

(3) 2

(4) 3

Question 4

The mean of 15 terms of a series consisting of positive integers is 13. What is the maximum possible value of the largest of the numbers?

(1) 180

(2) 90

(3) 100

(4) 80

Question 5

The sum of 10 distinct natural numbers is S. M is the maximum of those numbers, and is an integer. What would be the minimum value of S/M ?

(1) 4

(2) 3

(3) 2

(4) 1

Question 6

If $a + b + c = 10$, where a, b, and c are non-negative real numbers. Find the maximum value of abc.

Question 7

If $a + b + c = 10$, find the maximum value of $(a - 1)(b - 2)(c - 5)$. Given that $a \geq 1$, $b \geq 2$ and $c \geq 5$.

Question 8

If $a + b + c = 6$, find the maximum value of a^3b^2c, where all a, b and c are positive real numbers.

Question 9

If $x + y + z + w = 29$, where x, y, z and w are real numbers greater than 2, then find the maximum possible value of $(x - 1)(y + 3)(z - 1)(w - 2)$.

(1) 625

(2) 256

(3) 1296

(4) 2401

Question 10

If x, y and z are distinct positive integers, and $x + y + z = 11$, then the maximum value of $(xyz + xy + yz + zx)$ is

(1) 84

(2) 78

(3) 72

(4) 58

Question 11

p, q, r, s are any four positive real numbers, the minimum possible value of $\frac{p}{q}+\frac{q}{r}+\frac{r}{s}+\frac{s}{p}$ is

(1) 1

(2) 2

(3) $2\sqrt{2}$

(4) 4

Question 12

If a, b, c and d are four positive real numbers such that abcd = 1, what is the minimum value of; (1 + a) (1 + b) (1 + c) (1 + d)?

(1) 4

(2) 1

(3) 16

(4) 18

Question 13

Let x, y be two positive numbers such that x + y = 1. Then, the minimum value of

$$\left(x+\frac{1}{x}\right)^2+\left(y+\frac{1}{y}\right)^2 \text{ is}$$

(1) 12

(2) 20

(3) 12.5

(4) 13.3

Question 14

Let f(x) = max (2x + 1, 3 – 4x), where x is any real number. Then the minimum possible value of (x) is

(1) 1/3

(2) 1/2

(3) 2/3

(4) 5/3

Question 15

If the positive real numbers a, b and c are in Arithmetic Progression, such that abc = 4, then minimum possible value of b is

(1) $2^{\frac{3}{2}}$

(2) $2^{\frac{2}{3}}$

(3) $2^{\frac{1}{3}}$

(4) None of these

Question 16

From four positive real numbers a, b, c and d, 4 combinations of three numbers are chosen such that their sums are S_1, S_2, S_3 and S_4. If $a \times b \times c \times d = 5$, then find the minimum value of the product of $S_1 \times S_2 \times S_3 \times S_4$.

(1) 40

(2) 135

(3) 405

(4) 1080

Question 17

Davji Shop sells samosas in boxes of different sizes. The samosas are priced at Rs. 2 per samosa up to 200 samosas. For every additional 20 samosas, the price of the whole lot goes down by 10 paise per samosa. What should be the maximum size of the box that would maximise the revenue?

(1) 240

(2) 300

(3) 400

(4) 250

Question 18

If $\sqrt{x}-\sqrt{1-x}=\frac{1}{5}$ (x is a real number), then find the value of $\sqrt{x}+\sqrt{1-x}$.

(1) 2/5

(2) 3/5

(3) 6/5

(4) 7/5

Question 19

If pqr = 1, the value of the expression $\frac{1}{1+p+q^{-1}}+\frac{1}{1+q+r^{-1}}+\frac{1}{1+r+p^{-1}}$ is equal to

(1) p + q + r

(2) $\frac{1}{p+q+r}$

(3) 1

(4) $p^{-1}+q^{-1}+r^{-1}$

Question 20

How many pairs of positive integers m and n satisfy $\frac{1}{m}+\frac{4}{n}=\frac{1}{12}$, where n is an odd integer less than 60?

(1) 6

(2) 4

(3) 7

(4) 3

Question 21

The total number of integers pairs (x, y) satisfying the equation $x + y = xy$ is

(1) 0

(2) 1

(3) 2

(4) None of these

Question 22

If a, b, c, and d are integers such that $a + b + c + d = 30$, then the minimum possible value of $(a - b)^2 + (a - c)^2 + (a - d)^2$ is

Question 23

If a, b and c are positive integers such that $ab = 432$, $bc = 96$ and $c < 9$, then the smallest possible value of $a + b + c$ is

(1) 46

(2) 56

(3) 49

(4) 59

Question 24

If x and y are positive real numbers satisfying x + y = 102, then the minimum possible value of $2601\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)$ is

Question 25

For natural numbers x, y, and z, if $xy + yz = 19$ and $yz + xz = 51$, then the minimum possible value of xyz is

Question 26

If $c = \frac{16x}{y} + \frac{49y}{x}$ for some non-zero real numbers x and y, then c cannot take the value

(1) – 70

(2) – 60

(3) – 50

(4) 60

Question 27

Let $f(x) = \frac{x^2+1}{x^2-1}$ if $x \neq 1, -1$, and 1 if $x = 1, -1$. Let $g(x) = \frac{x+1}{x-1}$ if $x \neq 1$, and 3 if $x = 1$. What is the minimum possible value of $\frac{f(x)}{g(x)}$?

(1) 1/2

(2) 1/4

(3) –1

(4) 1/3

Question 28

Let $f(x) = \min \{2x^2, 5^2 - 5x\}$, where x is any positive real number. Then the maximum possible value of f(x) is

For Questions 29 and 30

Answer the questions on the basis of the information given below.

Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of + 5% while a fall will entail a return of –3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of –2.5%, while a fall will entail a return of +2%.

Question 29

The maximum guaranteed return to Shabnam is

(1) 0.25%

(2) 0.10%

(3) 0.20%

(4) 0.15%

Question 30

What strategy will maximize the guaranteed return to Shabnam?

(1) 100% in option A

(2) 36% in option B and 64% in option C

(3) 64% in option B and 36% in option C

(4) 1/3 in each of the three options

Thanks!